PART 1

10

EXPERIMENT 10

Determination of the Focal Length of a Convex Lens

Aim

To determine the focal length of a convex lens by obtaining an image of a distant object.

Requirements

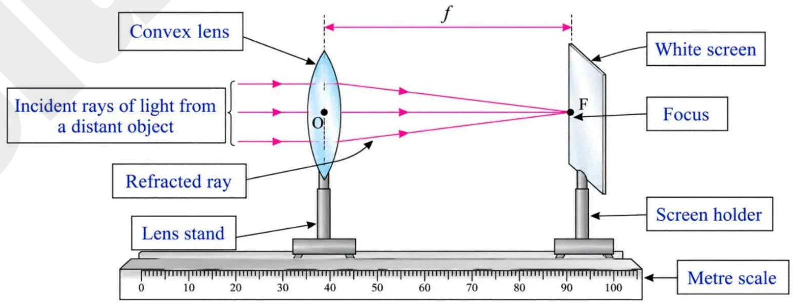
A convex lens, a lens stand, a metre scale, a white screen and a screen holder.

Procedure

1. Set up the convex lens on the lens stand.
2. Choose a distant object, such as an electric pole, and turn one surface of the lens towards it.
3. Set up a white screen in the screen holder and place it in front of the lens.
4. Move the screen closer to or further away from the lens until a sharp image of the pole is formed on it.
5. Measure the distance between the lens and the screen.
6. Repeat this procedure by choosing two more distant objects, such as a tree, a house, or a telephone pole.
7. Note down your readings in the observation table and work out the average focal length of the lens. Call this f_1.
8. Now turn the lens around by 180° so that its other surface faces the distant object, and obtain a sharp image of the object on the screen.
9. Measure the distance between the lens and the screen once again and record it. Take two more readings by choosing two more distant objects, and work out the average focal length. Call this f_2.
10. Using f_1 and f_2, work out the overall average focal length of the lens.
11. Compare f_1 and f_2 to check whether the lens is symmetric or not.

Diagram

(Label the various components in the given diagram)



Determination of the focal length of a convex lens

Observation Table

Sr. No.	Distant Object	Distance between the lens and screen (cm) — for one surface of the lens towards the object	Distance between the lens and screen (cm) — for the other surface of the lens towards the object	Average focal length of the lens $f = (f_1+f_2)/2$ (cm)
1	Electric pole	19.8	19.8	19.8
2	Tree	19.9	19.8	19.8
3	Window	19.7	19.7	19.7
		Average f_1 = 19.8	Average f_2 = 19.8	19.8

Inferences / Conclusions

1. The average focal length of the lens (f) = 19.8 cm.
2. $f_1 = f_2$. (Use proper sign, =, ≠)
3. The lens used is symmetric / ~~is not symmetric~~.

Precautions

1. Obtain the sharp image of a distant object on the screen.
2. Measure the distances accurately.

Multiple Choice Questions

1. The image formed by a concave lens is always ....

(A) real, inverted and diminished (B) real, inverted and magnified
(C) virtual, erect and magnified (D) virtual, erect and diminished

Answer: (D) virtual, erect and diminished

2. By using a ...., light rays incident on it can be converged.

(A) plane mirror (B) convex mirror
(C) concave lens (D) convex lens

Answer: (D) convex lens

3. The incident ray parallel to the principal axis of a convex lens passes through .... after refraction.

(A) the optical centre of the lens (B) focus F_2

(C) $2F_2$ (D) none of these

Answer: (B) focus F_2

Remember

The image obtained while finding the focal length of a convex lens is real and inverted. A convex lens is a converging lens.

Date: _______________ Teacher's Signature: _______________